BFS vs. extreme point

- Basic feasible solution (BFS)
  - Let A be an $m \times n$ matrix, $x \in \mathbb{R}^n$, and $b \in \mathbb{R}^m$. We consider the following system of linear equations
$$Ax = b \qquad (*)$$
We may assume without loss of generality that:
$$\text{rank}(A) = m < n. \qquad \underline{\text{(Full rank assumption)}}$$
Let B be an $m \times m$ submatrix consisting of m columns of A. If B is invertible, then B is called a <u>basis matrix</u>. The columns of A that are in B are called <u>basic columns</u>. The other columns of A are called <u>nonbasic columns</u>. The components of x corresponding to basic columns are called <u>basic variables</u>. The other components of x are called <u>nonbasic variables</u>. If x is a solution to $(*)$ with all nonbasic variables being 0, then x is said to be a <u>basic solution</u> to $(*)$ with respect to the basis matrix B.

  - eg Consider the linear system
$$\begin{aligned} 2x_1 + x_2 + x_3 + x_4 &= 15 \\ x_1 + 3x_2 + x_3 - x_5 &= 12 \end{aligned} \qquad \text{i.e.,} \quad \begin{bmatrix} 2 & 1 & 1 & 1 & 0 \\ 1 & 3 & 1 & 0 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 15 \\ 12 \end{bmatrix}$$
We can choose $B = \begin{bmatrix} A_1 & A_3 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$, then
basic variables: $x_B = \begin{bmatrix} x_1 \\ x_3 \end{bmatrix} = B^{-1}b = \begin{bmatrix} 3 \\ 9 \end{bmatrix}$, nonbasic variables: $x_N = \begin{bmatrix} x_2 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

$x = (3, 0, 9, 0, 0)$ is a basic solution of the linear system

If we choose $B = \begin{bmatrix} A_4 & A_1 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$ instead, then
basic variables: $x_B = \begin{bmatrix} x_4 \\ x_1 \end{bmatrix} = B^{-1}b = \begin{bmatrix} -9 \\ 12 \end{bmatrix}$, nonbasic variables: $x_N = \begin{bmatrix} x_2 \\ x_3 \\ x_5 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

$x = (12, 0, 0, -9, 0)$ is a basic solution of the linear system.

- Remark: 1. In a basic solution, the nonbasic variables are all zero by definition. The basic variables are not necessarily nonzero. In fact, we have the following definition:

Def: If one or more of the basic variables in a basic solution has value zero, that solution is called a <u>degenerate basic solution</u>.

2. The full rank assumption can be made without loss of generality.

- If $\text{rank}(A) < m$, then there are either contradictory constraints or redundant constraints.

e.g
$$\begin{aligned} 2x_1 + x_2 + x_3 &= d \\ x_1 + x_2 \quad\quad &= 1 \\ x_1 \quad\quad + x_3 &= 1 \end{aligned} \qquad A = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} \quad \text{rank}(A) = 2 < 3$$

The system has no solution if $d \neq 2$. The first constraint is redundent if $d = 2$.

- If $\text{rank}(A) = m = n$, then the set $\{x : Ax = b,\ x \geq 0\}$ contains at most one point. There's nothing to optimize.

- Now we consider the feasible region of a linear program in standard form.

$$S = \{x : Ax = b,\ x \geq 0\}.$$

Def: Let x be a basic solution to the linear system $Ax = b$. If $x \geq 0$, (i.e., x is a feasible solution,) then x is called a <u>basic feasible solution</u>. If x is also degenerate, we call x a <u>degenerate basic feasible solution</u>.

e.g.: In the previous example, $x = (3, 0, 9, 0, 0)$ is a basic feasible solution, but $x = (12, 0, 0, -9, 0)$ is not feasible. They are both nondegenerate.

· Equivalence of BFS and extreme points.

Thm Let $S = \{x \; Ax = b, x \geq 0\}$, where A is an $m \times n$ matrix and satisfies the full rank assumption.

x is an extreme point of S if and only if x is a BFS to $\begin{cases} Ax = b \\ x \geq 0 \end{cases}$.

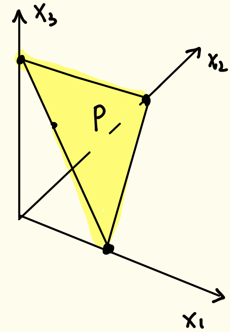
e.g. Consider the set in $\mathbb{R}^3$ defined by

$$x_1 + x_2 + x_3 = 1,$$
$$x_1, x_2, x_3 \geq 0$$

BFS: $B = [A_1]$: $x = (1, 0, 0)$

$B = [A_2]$: $x = (0, 1, 0)$

$B = [A_3]$: $x = (0, 0, 1)$



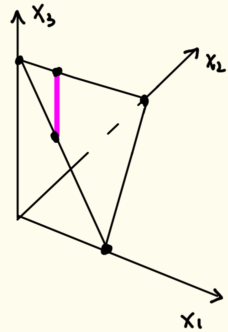
e.g.: Consider the set in $\mathbb{R}^3$ defined by

$$x_1 + x_2 + x_3 = 1,$$
$$2x_1 + 3x_2 \quad = 1,$$
$$x_1, x_2, x_3 \geq 0$$

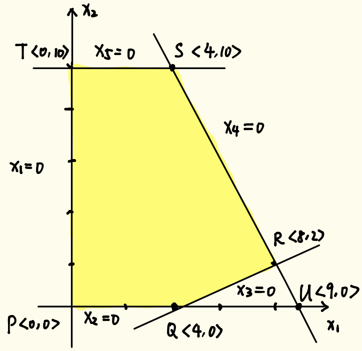
BFS: $B = [A_1 \; A_2]$: $x = (2, -1, 0)$ X infeasible

$B = [A_1, A_3]$: $x = (1/2, 0, 1/2)$

$B = [A_2, A_3]$: $x = (0, 1/3, 2/3)$



eg Consider the set in $\mathbb{R}^2$ defined by



$$
\begin{aligned}
x_1 - 2x_2 &\le 4 \\
2x_1 + x_2 &\le 18 \\
x_2 &\le 10 \\
x_1, x_2 &\ge 0
\end{aligned}
$$

In order to compare this example with our results, we introduce slack variables and consider the equivalent set in $\mathbb{R}^5$ defined by:

$$
\begin{aligned}
x_1 - 2x_2 + x_3 \quad\quad\quad &= 4 \\
2x_1 + x_2 \quad + x_4 \quad &= 18 \\
x_2 \quad\quad + x_5 &= 10 \\
x_1, x_2, x_3, x_4, x_5 &\ge 0
\end{aligned}
$$

① Choose $B = [A_2 \ A_3 \ A_4] = \begin{bmatrix} -2 & 1 & 0 \\ 1 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$, invertible

$$x_N = \begin{bmatrix} x_1 \\ x_5 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$Bx_B = b \Rightarrow \begin{bmatrix} -2 & 1 & 0 \\ 1 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 4 \\ 18 \\ 10 \end{bmatrix} \Rightarrow x_B = \begin{bmatrix} x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 10 \\ 24 \\ 8 \end{bmatrix}$$

$x = (0, 10, 24, 8, 0)$ is a basic solution

Since $x_B \ge 0$, x is a basic feasible solution

On the graph, T is an extreme point

② Choose $B = [A_1\ A_3\ A_5] = \begin{bmatrix} 1 & 1 & 0 \\ 2 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ is invertible

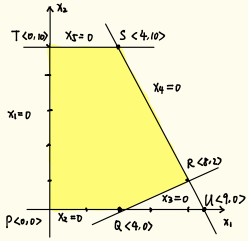
$$x_N = \begin{bmatrix} x_2 \\ x_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$Bx_B = b \Rightarrow \begin{bmatrix} 1 & 1 & 0 \\ 2 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_3 \\ x_5 \end{bmatrix} = \begin{bmatrix} 4 \\ 18 \\ 10 \end{bmatrix} \Rightarrow x_B = \begin{bmatrix} x_1 \\ x_3 \\ x_5 \end{bmatrix} = \begin{bmatrix} 9 \\ -5 \\ 10 \end{bmatrix}$$

$x = (9, 0, -5, 0, 10)$ is a basic solution

Since $x_B \not\geq 0$, x is NOT a basic feasible solution

On the graph, U is NOT an extreme point



Remark: 1. $B = [A_1\ A_3\ A_4] = \begin{bmatrix} 1 & 1 & 0 \\ 2 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$ is NOT invertible

B cannot be a basis matrix

2. There is a one-to-one correspondence between

$\{x : Ax \leq b, x \geq 0\}$

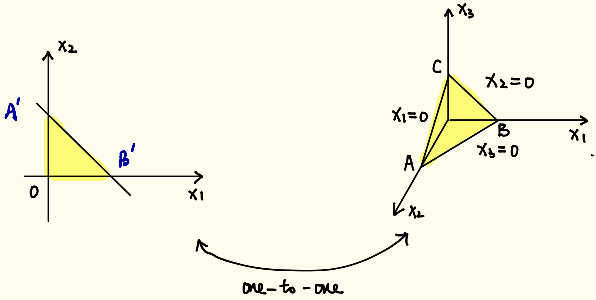
and

$\{(x, s) : Ax + s = b,\ x, s \geq 0\}$

eg $x_1 + x_2 \leq 1$
$x_1 \geq 0$
$x_2 \geq 0$

$\xrightarrow{\text{standard form}}$

$x_1 + x_2 + x_3 = 1$
$x_1, x_2, x_3 \geq 0$



- Fundamental theorem of linear programming

Thm: Consider a linear program in standard form,

$$\begin{aligned} \max \quad & c^T x \\ \text{s.t.} \quad & Ax = b \\ & x \geq 0 \end{aligned}$$

Under the full rank assumption of $A_{m \times n}$,

i) if there is a feasible solution, there is a BFS.

ii) if there is an optimal solution, there is an optimal BFS.